

Graceful labelings of trees with a degree-4 and a degree-3 branch vertex

Machine-checked in Lean 4 · github.com/dannyteegus/graceful-10 ·
doi:10.5281/zenodo.23248513

Daniel Taylor

8 October 2026

Theorem. Every finite tree with exactly two branch vertices, one of degree 4 and one of degree 3, is graceful.

Status: proved and machine-checked. `proof/Main.lean` passes conjectures `verify` for task `fc-4b69a7dc-math15catalog-source10-a324196efa-formalized-v1` (accepted, reason VERIFIED, exit 0, 2026-10-08, local development sandbox).

The proof is constructive. It builds explicit labelings from a small toolkit of α -labeling operations, plus a finite number of explicit labelings found by computer and checked in Lean.

0. Setting and notation

A *graceful labeling* of a tree T with m edges is an injection $f : V(T) \rightarrow \{0, \dots, m\}$ such that the edge labels $|f(x) - f(y)|$ over the edges xy are exactly $\{1, \dots, m\}$.

Every tree with exactly two branch vertices, of degrees 4 and 3, is isomorphic to

$$T = T(a, b, c \mid d \mid e, f), \quad a, b, c, d, e, f \geq 1,$$

which consists of:

- a vertex u (degree 4) with pendant paths (“arms”) of a, b, c vertices;
- a vertex v (degree 3) with arms of e, f vertices;
- a u - v path with d edges (the “ d -path”); its interior vertices are $d_1, \dots, d_{d-1}$, where d_i is at distance i from u .

By symmetry we may assume $a \leq b \leq c$ and $e \leq f$. Arm vertices are written x_k (the k -th vertex of arm x , counted from its root).

α -labelings. An α -labeling is a graceful labeling with a threshold λ such that every edge joins a label $\leq \lambda$ (“low”) to a label $> \lambda$ (“high”). The low vertices form one colour class of the bipartition. For a vertex x define:

- $\varepsilon(x) = f(x)$ if x is low, and $m - f(x)$ if x is high (distance to the extreme label);

- $\tau(x) = \lambda - f(x)$ if low, and $f(x) - \lambda - 1$ if high (distance to the threshold).

Then $\varepsilon(x) + \tau(x)$ is constant on each colour class.

Notation. $\text{OK}(n, t)$ means: $n \geq 1$, $0 \leq t \leq \lfloor (n-1)/2 \rfloor$, and **not** $(n \equiv 1 \pmod{4})$, $n > 1$, $4t = n - 1$.

1. Tools

Lemma 1.1 (complement). If f is graceful then so is $m - f$. If f is α , so is $m - f$, with ε and τ unchanged.

Lemma 1.2 (reflection). If f is α with threshold λ , define $g(x) = \lambda - f(x)$ for low x and $g(x) = m + \lambda + 1 - f(x)$ for high x . Then g is α with the same threshold and the same low set, and g swaps ε and τ at every vertex. (An edge label $h - l$ becomes $m + 1 - (h - l)$.)

Lemma 1.3 (gluing). Let A be α -labeled (m_A edges, threshold λ), $x \in A$, and let B be gracefully labeled by g (m_B edges), $w \in B$. Join them by the edge xw . Put $F = f_A$ on the low part of A , $F = f_A + m_B + 1$ on the high part, and $F = \lambda + 1 + g$ on B . Then F is graceful iff $g(w) = \tau(x)$ (if x is high) or $g(w) = m_B - \tau(x)$ (if x is low). If moreover g is α and w lies in the colour class opposite to x , then F is α , ε is unchanged on A , and τ is unchanged on B .

Lemma 1.4 (end invariant). In any α -labeling of a path $z_1 \dots z_n$ with z_1 low, $\tau(z_n) = f(z_1)$.

Theorem 1.5 (α -EPL). The path on n vertices has an α -labeling with threshold $\lfloor (n-1)/2 \rfloor$ whose first vertex is low with label t whenever $\text{OK}(n, t)$ holds. (Brute force shows the excluded cases are genuine for $n \leq 14$; this is not used.)

Construction, by strong induction on n :

- $t = 0$: the zigzag $0, m, 1, m-1, \dots$
- $2t > \lambda$: reflect a labeling for $\lambda - t$.
- $2t = \lambda$: pattern C, $t, 3t, t+1, 3t-1, \dots, 2t, m, 0, m-1, 1, \dots, m-t$.
- $2t < \lambda$: pattern E or E2. A zigzag head ends at an extreme label, and a smaller α -EPL instance is shifted into the remaining box.

Corollary 1.6 (attaching an α segment). Let A be α , $x \in A$, and attach at x a new path of L vertices with $\text{OK}(L, \tau(x))$. Then (Lemma 1.3 with Theorem 1.5) the result A' is α . Moreover:

- ε is unchanged on the old vertices;
- for every old z , $\tau(z)$ grows by $\lfloor (L+\delta)/2 \rfloor$, where $\delta = 1$ iff z and x lie in different colour classes;
- the far end of the new path has $\tau = \tau_A(x)$, by Lemma 1.4.

Corollary 1.7 (final segment). If A is α , $x \in A$ and $\tau(x) \leq L-1$, then attaching a path of L vertices at x gives a graceful labeling G with $G(z) \in \{\varepsilon(z), m - \varepsilon(z)\}$ for every old z . (Glue a graceful path whose end has label $\tau(x)$.)

Lemma 1.8 (leaf chain). If G is graceful and $G(x) \in \{0, m\}$, then attaching a path of $L \geq 0$ new vertices at x keeps the tree graceful. (Give each new leaf the label

$m + 1$, then complement.)

Lemma 1.9 (join). Let A be α , B graceful, and suppose the only tree edge between them is xw . If $B(w) \in \{\tau_A(x), m_B - \tau_A(x)\}$ and $\tau_A(x) \leq m_B$, then the union is graceful (Lemma 1.3, complementing B if needed).

2. Pieces

Below, “ X at r ” names an arm X hanging at the vertex r . A *program* is a sequence of the operations of §1: start with an explicit α -path, attach α segments (Cor. 1.6), reflect (Lemma 1.2), and finish with a final segment (Cor. 1.7) or a join (Lemma 1.9). Each program’s side conditions are $\text{OK}(\cdot, \cdot)$ constraints, computed by the ε/τ bookkeeping of Cor. 1.6.

Lemma 2.1 (IVL0). For all $(x, y) \neq (2, 2)$, the path $P(x, y)$ (arms x, y at a root r) has an α -labeling with $\varepsilon(r) = 0$ and $\tau(r) = \lfloor x/2 \rfloor + \lfloor y/2 \rfloor$. Assume $x \leq y$:

- (a) zigzag along x from r , then attach y at r ; needs $\text{OK}(y, \lfloor x/2 \rfloor)$;
- the explicit core for $(6, 6)$;
- (b) x_1, y_1 at r , then the two tails; for $x \geq 4, x \neq 6$;
- the programs $s1$ (for $(3, 5)$) and $s0$ (for $(2, 5), (6, 13)$) on the swapped pair.

A short case analysis shows that every pair is covered.

Lemma 2.2 (SP0). For every $a \leq b \leq c$, the spider $S(a, b, c)$ has a graceful labeling with the centre u at label 0 (or m):

- generic: $\text{IVL0}(a, b)$ at u , then a final segment c ; needs $\lfloor a/2 \rfloor + \lfloor b/2 \rfloor \leq c - 1$, which holds unless $a = b = c$ is even;
- $(2, 2, c)$: for $c \geq 9$, an explicit α -core $S(2, 2, 5)$ plus a final segment; for $c \leq 8$, tables;
- $(2k, 2k, 2k)$: for $k \geq 8$, explicit α -cores ($S(7, 1, 1)$, or $S(7, 3, 1)$ when $k = 12$) plus α tails and a final segment; for $k \leq 7$, tables.

Lemma 2.3 (SP σ , $\sigma = 1, 2$). A graceful spider at u with u at label σ or $m - \sigma$ can be built by any of these programs on an ordering (x, y, z) of the arms:

- spA : α -EPL from u along x with u at σ , then y at u , then a final z . Needs $\text{OK}(x + 1, \sigma)$, $\text{OK}(y, \lfloor x/2 \rfloor - \sigma)$, and $\lfloor x/2 \rfloor - \sigma + \lfloor y/2 \rfloor \leq z - 1$.
- sw1 ($\sigma = 1$): $\text{IVL0}(x, y)$, reflect, $z_1 z_2$, reflect, final rest of z . Needs $(x, y) \neq (2, 2)$ and $\lfloor x/2 \rfloor + \lfloor y/2 \rfloor + 4 \leq z$.
- sw2 ($\sigma = 1$): zigzag x , reflect, $y_1 y_2$, reflect, y tail, final z . Needs $y \geq 3$, $\text{OK}(y - 2, \lfloor x/2 \rfloor + 1)$, and $\lfloor x/2 \rfloor + \lfloor y/2 \rfloor \leq z$.

Lemma 2.4 (SP1 coverage). Every $a \leq b \leq c$ except the 13 spiders

$$F_1 = \{(1, 1, 1), (1, 4, 4), (4, 4, 4), (4, 4, 6), (4, 4, 7), (4, 6, 6), (4, 6, 7), \\ (4, 6, 8), (4, 7, 7), (4, 7, 8), (4, 8, 8), (5, 5, 5), (6, 9, 9)\}$$

has an SP1 program:

- spA on (a, b, c) whenever $a \geq 2$, $a \neq 4$, and b is not exceptional, where exceptional means $b \equiv 1 \pmod{4}$ and $b = 4\lfloor a/2 \rfloor - 3$;
- $a = 1$: by $b - \text{sw1}$, spA(c, a, b), spA(b, a, c), sw2(a, b, c), or sw2(a, c, b);
- $a = 4$: sw2 for $b \geq 9$, $b \neq 15$; otherwise sw1 when $c \geq \lfloor b/2 \rfloor + 6$;
- exceptional b : sw1 when $c \geq \lfloor a/2 \rfloor + \lfloor b/2 \rfloor + 4$;
- the 12 remaining spiders, (4,4,5), (4,5,5-7), (4,8,9), (5,5,6-7), (6,9,10), (7,9,9-10), (8,13,13) and (9,13,13): each gets an explicit choice of program.

Lemma 2.5 (v-side pieces, $e = f = 2$). Let $\text{VSP}(D, \sigma)$ be an α -labeling of the subtree made of v , its arms of length 2, and the D path vertices $d_{d-1}, \dots, d_{d-D}$, with $\tau = \sigma$ at its last vertex (v itself when $D = 0$). It exists:

- for $\sigma = 1$: when $D \in \{0, 1, 5\}$ or $\text{OK}(D, 1)$, i.e. for every $D \neq 2$ (graceful $P(2, 2) = (0, 4, 1, 3, 2)$ plus a chain, or explicit cores);
- for $\sigma = 2$: when $D \in \{3, 4\}$, or $D > 3$ and $\text{OK}(D - 3, 2)$, or $D > 4$ and $\text{OK}(D - 4, 2)$ (explicit cores plus a chain). In particular for $D \in \{3, 4\} \cup [8, \infty)$.

3. Family A: $(e, f) \neq (2, 2)$

Take $\text{IVL0}(e, f)$ at v and reflect, so $\tau(v) = 0$. If $d \geq 2$, attach the $d - 1$ interior d -path vertices at v ; this is allowed since $\text{OK}(d - 1, 0)$ always holds, and the new end d_1 has $\tau = 0$. Join with $\text{SP0}(a, b, c)$ at u (Lemma 1.9, condition $u \mapsto 0$ or m).

4. Family C: $e = f = 2$

There are five templates.

- **V(σ)**: $\text{VSP}(d - 1, \sigma)$ joined to an $\text{SP}\sigma$ spider at the edge $d_1 u$ (Lemma 1.9).
- **cut**: $\text{VSP}(d - 2, 1)$, joined at $d_2 d_1$ to a graceful piece containing d_1, u and the spider with d_1 at label 1. That piece is: the α -path d_1, u, x_1 labelled 1, 2, 0; the α tail of x ; αy at u ; and a final z . It needs $\text{OK}(x - 1, 1)$, $\text{OK}(y, \lfloor x/2 \rfloor)$ and $\lfloor x/2 \rfloor + \lfloor y/2 \rfloor \leq z - 1$.
- **U**: an α piece on the u -side, joined to a graceful $P(2, 2)$ at v . The α piece is: zigzag x from u ; the d -path from u ; reflect; αy and z at u ; reflect. Its junction d_{d-1} has $\tau = \lfloor x/2 \rfloor$, and the $P(2, 2)$ has v at ρ or $4 - \rho$ for $\rho \in \{0, 1, 3, 4\}$. It needs $\lfloor x/2 \rfloor \in \{0, 1, 3, 4\}$, $d = 1$ or $\text{OK}(d - 1, \lfloor x/2 \rfloor)$, and $\text{OK}(z, \lfloor y/2 \rfloor)$.
- **Π ($d = 3$)**: an explicit α -path $f_2 f_1 v d_2 d_1 u (a_1 \dots)$ with v at label 0, then αb at u , a final c at u , and a leaf chain for arm e at v (Lemma 1.8). The starts are:

a	start labels	$\tau(u)$
1	1, 5, 0, 6, 3, 4, 2	0
2	1, 6, 0, 7, 3, 4, 2, 5	0
3	1, 8, 0, 6, 4, 5, 2, 7, 3	0
4	1, 8, 0, 9, 3, 5, 4, 7, 2, 6	0
$a \geq 5, a \neq 7$	the $a = 2$ start plus an α tail at a_2	$\lfloor (a - 2)/2 \rfloor$
Π'	$f_2 \dots u$ labelled 1, 5, 0, 3, 2, 4 plus αa at u	$1 + \lfloor a/2 \rfloor$

- **cores:** explicit graceful labelings with u at 0, extended by a leaf chain at u (Lemma 1.8):
 - $T(4, 4, \cdot \mid d \mid 2, 2)$ and $T(\cdot, 4, 4 \mid d \mid 2, 2)$ for $d \in \{1, 2, 6, 7, 8\}$;
 - $T(6, 9, \cdot \mid d \mid 2, 2)$ for $d \in \{2, 6, 7\}$;
 - plus one table, $T(5, 5, 5 \mid 1 \mid 2, 2)$.

Coverage.

- $d = 3$, any $a \leq b \leq c$. Use Π for $a \in \{1, 2, 3, 4\}$, and Π for $a \geq 5$, $a \neq 7$, unless $b \equiv 1 \pmod{4}$ and $4 \lfloor (a-2)/2 \rfloor = b-1$. Otherwise use cut, or Π' when $(a, b) = (7, 13)$ or $a = 6$.
- $d \neq 3$, **spider not in** F_1 : $V(1)$. $VSP(d-1, 1)$ exists since $d-1 \neq 2$, and SP1 exists by Lemma 2.4.
- $d \neq 3$, **spider in** F_1 :
 - $(1, 1, 1)$: U.
 - $(1, 4, 4)$ and $(4, 4, c)$: cores for $d \in \{1, 2, 6, 7, 8\}$, $V(2)$ otherwise.
 - $(4, b, c)$ with $6 \leq b \leq c \leq 8$: U for $d = 1$, $V(2)$ for $d = 4$, cut otherwise.
 - $(5, 5, 5)$: the table for $d = 1$, $V(2)$ for $d = 4$, cut otherwise.
 - $(6, 9, 9)$: U for $d \in \{1, 8\}$, cores for $d \in \{2, 6, 7\}$, $V(2)$ otherwise.

5. Formalization

The whole argument is formalized in Lean 4 / Mathlib in `proof/Main.lean`, which concatenates `proof/A1...F8`, as theorem `target : Math15.Graceful.Target10`.

- **Structure theorem** (C1–C9): every tree with branch vertices of degrees 4 and 3 is isomorphic to a canonical $T(a, b, c \mid d \mid e, f)$ on natural-number names ($u = 0$, $v = 1$, arm X position $k = X + 8k$). This reduces the target to `CanonGraceful a b c d e f` for sorted parameters.
- **α -piece API** (D1–E1): pieces carry their α threshold and the ε/τ bookkeeping. `Attach`, `final-segment`, `join` and `leaf-chain` lemmas are proved once, with symbolic arm lengths.
- **Explicit labelings** are checked by `decide +kernel` on a Boolean tree-and-labeling checker. `native_decide` is not used.
- **Coverage** (F1–F8) is proved by omega case splits that mirror §3–§4.

Note. This proof was produced with AI coding agents; the author directed the work and verified the result with the `conjectures.io` checker.